

The ideal gas law

$$PV = KT$$

Where P is the pressure, V the volume, T the absolute temp of gas and K is constant. We can rearrange the equation

$$P = \frac{KT}{V}$$

This shows that P is a function of T and V . If one or the other say T is kept fixed and V changes, the derivative of P w.r.t.

V measures the rate of change of pressure w.r.t volume. In this case, it is called the partial derivative of P with respect to V and written as.

$$\frac{\partial P}{\partial V}$$

Example 1 If $P = \frac{KT}{V}$ find the partial derivatives of P
(a) with respect to T , (b) with respect to V .

Solution

$$(a) \quad P = \frac{KT}{V}$$

$$\frac{\partial P}{\partial T} = \frac{K}{V}$$

Where V is treated as a constant for the calculation

b) for this part, T is treated as a constant. Thus

$$P = KT \frac{1}{V} = KT V^{-1}$$

$$\frac{\partial P}{\partial V} = -KT V^{-2} = -\frac{KT}{V^2}$$

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symbol ∂ is used (unlike $\frac{d}{dx}$ where $\frac{d}{dx}$ is used) but the techniques of partial differentiation is being differentiated but the techniques of partial differentiation are exactly the same as for (ordinary) differentiation.

② If $z = 5x^4 + 2x^3y^2 - 3y$ find

$$(a) \frac{dz}{dx} \quad \text{and} \quad (b) \frac{dz}{dy}$$

(a) To find $\frac{dz}{dx}$, y is kept constant

$$\text{Since } z = 5x^4 + (2y^2)x^3 - (3y)$$

$$\lim \frac{\partial z}{\partial x} = \frac{d}{dx}(5x^4) + (2y^2) \frac{d}{dx}(x^3) - 3y \frac{d}{dx}(1)$$

$$= 20x^3 + 2y^2 \cdot 3x^2 - 0$$

$$\text{Hence } \frac{\partial z}{\partial x} = 20x^3 + 6y^2x^2$$

b) To find $\frac{\partial z}{\partial y}$, x is kept

$$\text{Since } z = (5x^4) + (2x^3)y^2 - 3y$$

$$\lim \frac{\partial z}{\partial y} = (5x^4) \frac{\partial}{\partial y}(1) + (2x^3) \frac{\partial}{\partial y}(y^2) - 3 \frac{\partial}{\partial y}(y)$$

$$= 0 + 4x^3y - 3$$

$$\frac{\partial z}{\partial y} = 4x^3y - 3$$

Q.1 $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for the equation

$$Z = (x^4 + x^3)y^3$$

∂z to calculate $\frac{\partial z}{\partial x}$, y is treated as a constant.

$$\begin{aligned}\frac{\partial z}{\partial x} &= \frac{\partial}{\partial x} ((x^4 + x^3)y^3) \\ &= \frac{\partial}{\partial x} (x^4 + x^3) * y^3 \\ &= 4x^3 + 3x^2)y^3\end{aligned}$$

To find the partial derivative $\frac{\partial z}{\partial y}$ the factor $(x^4 + x^3)$ is treated as a constant.

$$\begin{aligned}\frac{\partial z}{\partial y} &= \frac{\partial}{\partial y} ((x^4 + x^3)y^3) = (x^4 + x^3) * \frac{\partial}{\partial y} y^3 \\ \frac{\partial z}{\partial y} &= 3(x^4 + x^3)y^2\end{aligned}$$

EXAMPLE (4)

find the $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for the equation $Z = y^{1/2} \sin(x)$

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} (y^{1/2} \sin(x)) = y^{1/2} \frac{\partial}{\partial x} (\sin(x))$$

$$\frac{\partial z}{\partial x} = y^{1/2} \cos(x)$$

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y} (y^{1/2} \sin(x))$$

$$= \frac{\partial}{\partial y} y^{1/2} * \sin(x).$$

$$\frac{\partial z}{\partial y} = \frac{1}{2} y^{-1/2} \sin(x).$$

Product Rules of Partial Differentiation

Since partial differentiation is essentially the same as ordinary differentiation the product, quotient and chain rules may be applied

$$(a) \ z = xy \cos(xy) \quad (b) \ z = \frac{x-y}{x+xy}$$

$$(a) \ u = xy \quad \text{and} \quad v = \cos(xy)$$

$$\frac{du}{dx} = y \quad \text{and} \quad \frac{dv}{dx} = -y \sin(xy)$$

$$\text{Thus } \frac{\partial z}{\partial x} = \frac{\partial u}{\partial x} v + u \frac{\partial v}{\partial x} = y \cos(xy) - xy^2 \sin(xy)$$

$$b) \text{ Here } z = \frac{u}{v};$$

$$\text{where } u = x-y \text{ and } v = x+xy$$

$$\frac{\partial u}{\partial x} = 1 \quad ; \quad \frac{\partial v}{\partial x} = 1$$

$$\frac{\partial z}{\partial x} = \frac{v \frac{\partial u}{\partial x} - u \frac{\partial v}{\partial x}}{v^2}$$

$$= \frac{(x+y) - (x-y)}{(x+y)^2} = \frac{2y}{(x+y)^2}$$

$$Z = U^2$$

$$Z = (3x+y)^2 \quad \text{So}$$

$$\text{where } U = (3x+y)$$

Applying the chain rule.

$$\frac{\partial Z}{\partial x} = 2U \quad \text{and} \quad \frac{dU}{dx} = 3$$

$$\frac{\partial Z}{\partial x} = \frac{\partial Z}{\partial U} \cdot \frac{dU}{dx} \quad ; \quad \frac{\partial Z}{\partial y} = \frac{\partial Z}{\partial U} \cdot \frac{dU}{dy}$$

$$2(3x+y) \cdot 3 = 6(3x+y)$$

$$a) \quad Z = (X^2 + 3X) \sin y \quad \text{find} \quad \frac{\partial Z}{\partial x} \quad \text{and} \quad \frac{\partial Z}{\partial y}$$

$$\text{let } U = (X^2 + 3X)$$

$$V = \sin y$$

Apply the product rule

$$\frac{dU}{dx} = (2x+3) \quad ; \quad \frac{dV}{dx} = 0$$

$$\frac{dU}{dy} = 0 \quad ; \quad \frac{dV}{dy} = (\cos y)$$

Applying Product rule.

$$\frac{\partial Z}{\partial x} = \frac{\partial U}{\partial x} V + U \frac{\partial V}{\partial x} = (2x+3) \sin y$$

$$\frac{\partial Z}{\partial y} = V \frac{\partial U}{\partial y} + U \frac{\partial V}{\partial y} = (X^2 + 3X) (\cos y)$$

Pressure P of a mass of gas is given by $PV = mRT$,
 m and R are constants, V is the volume and T the temperature.
Find expressions for $\frac{\partial P}{\partial T}$ and $\frac{\partial P}{\partial V}$

Since $PV = mRT$ then $P = \frac{mRT}{V}$

To find $\frac{\partial P}{\partial T}$, V is kept constant.

$$\text{Hence } \frac{\partial P}{\partial T} = \left(\frac{mR}{V}\right) \frac{d}{dT}(T) = \frac{mR}{V}$$

To find $\frac{\partial P}{\partial V}$, T is kept constant

$$\begin{aligned}\text{Hence } \frac{\partial P}{\partial V} &= mRT \frac{d}{dV}\left(\frac{1}{V}\right) \\ &= mRT(-V^{-2}) = -\frac{mRT}{V^2}\end{aligned}$$

Problem 6. The time of oscillation, t , of a pendulum is given by
 $t = 2\pi \sqrt{l/g}$ where l is the length of the pendulum and g the
free fall acceleration due to gravity. Determine $\frac{dt}{dl}$ and $\frac{dt}{dg}$.

To find $\frac{dt}{dl}$, g is kept constant

$$t = 2\pi \sqrt{l/g} = \frac{2\pi}{\sqrt{g}} \sqrt{l} = \left(\frac{2\pi}{\sqrt{g}}\right) l^{1/2}$$

$$\text{Hence } \frac{dt}{dl} = \left(\frac{2\pi}{\sqrt{g}}\right) \frac{d}{dl}(l^{1/2}) = \left(\frac{2\pi}{\sqrt{g}}\right) \left(\frac{1}{2} l^{-1/2}\right)$$

$$= \left(\frac{2\pi}{\sqrt{g}}\right) \left(\frac{1}{2\sqrt{l}}\right) = \frac{\pi}{\sqrt{lg}}$$

To find $\frac{\partial t}{\partial g}$, l is kept constant

$$t = 2\pi\sqrt{\frac{l}{g}} = (2\pi\sqrt{l})\left(\frac{1}{\sqrt{g}}\right) \\ = (2\pi\sqrt{l})g^{-1/2}$$

$$\text{Hence } \frac{\partial t}{\partial g} = (2\pi\sqrt{l})\left(-\frac{1}{2}g^{-3/2}\right) \\ = (2\pi\sqrt{l})\left(\frac{-1}{2\sqrt{g^3}}\right) \Rightarrow (2\pi\sqrt{l})^{-1/2}\left(\frac{1}{\sqrt{g^3}}\right)$$

$$\frac{\partial t}{\partial g} = \frac{-\pi\sqrt{l}}{\sqrt{g^3}} = -\pi\sqrt{\frac{l}{g^3}}$$

SECOND ORDER PARTIAL DERIVATIVES.

(i) If $V = \pi r^2 h$.

$$\text{Then } \frac{\partial^2 V}{\partial r^2} = \frac{\partial}{\partial r}(2\pi r h) = 2\pi h = 2\pi h.$$

(ii) to determine $\frac{\partial^2 V}{\partial h^2}$ wrt h , keeping r constant, gives

$$\frac{\partial}{\partial h}\left(\frac{\partial V}{\partial h}\right) \text{ which is } V = \pi r^2 h$$

$$\frac{\partial^2 V}{\partial h^2} = \frac{\partial}{\partial h}(\pi r^2)$$

$$\text{Thus } \frac{\partial^2 V}{\partial h^2} = \frac{\partial}{\partial h}(\pi r^2) = 0.$$

Let a country's production can be described by the function

$$Y(L, K) = 15L^{1/3}K^{2/3}$$

its, where L is units of labour and K is unit of Capital

Calculate the marginal productivity of labour and the marginal productivity of Capital.

Evaluate Y_L and Y_K when $L = 100$ and $K = 55$.

Interpret your result.

The marginal productivity of labour is calculated to be.

$$Y_L = \frac{\partial}{\partial L} 15L^{1/3}K^{2/3} = 5\left(\frac{L}{K}\right)^{2/3}$$

and the marginal productivity of Capital is

$$Y_K = \frac{\partial}{\partial K} 15L^{1/3}K^{2/3} = 10\left(\frac{L}{K}\right)^{1/3}$$

b. When $L = 100$ and $K = 55$, we have

$$Y_L(100, 55) = 55\left(\frac{55}{100}\right)^{2/3} \approx 3.36$$

and

$$Y_K(100, 55) = 10\left(\frac{55}{100}\right)^{1/3} \approx 12.2$$